University of California, Berkeley *Prof. Jun-Chau Chien*
EE 240C Spring 2026 *GSI: Nikhil Jain*

EE240C ADC Project

In this project, you will design an ADC based on an architecture selected from the literature. The project will be divided into four phases. You will begin by developing a behavioral-level model of the ADC and performing both functional and performance verification. The behavioral model should ideally be implemented in Verilog-A and simulated using Cadence Spectre. This platform enables seamless co-simulation with transistor-level models, allowing individual blocks to be progressively replaced with transistor-level schematics for systematic debugging as the system is developed.

In the behavioral model, you are also expected to incorporate circuit non-idealities, such as noise, nonlinearity, bandwidth limitations, and offset. While it is not necessary to replace the entire ADC with transistor-level schematics, you must perform at least one co-simulation in which a key block, such as the comparator or amplifier, is replaced with its transistor-level implementation.

The suggested ADC architectures are noise-shaping SAR, delta-sigma modulator, and time-based ADC.

Phase I: Form teams of two students. Select a paper from ISSCC, VLSI Circuits, CICC, JSSC, or other IEEE conferences/journals, and perform an in-depth analysis of the chosen work. Prepare a midterm presentation (3–5 slides) to introduce the architecture to the class and outline your implementation plan. The class will provide feedback on your proposed approach.

Please include a summary table of key specifications from the selected paper, and clearly state your target specifications. For example, you may aim to achieve a 10× higher sampling rate while maintaining comparable performance in other metrics, or improve the resolution by 2 additional ENOB.

Deliverables (due: 3/31/2026): Please submit the following materials:

- Team information
- Paper title (to be entered in the shared Excel sheet)
- The PDF of the selected paper
- Presentation slides

Both the team information and paper title should be recorded in the shared Excel sheet for coordination and tracking.

Phase II: You will begin by developing the behavioral model of your chosen architecture. Start with ideal components, meaning that no non-idealities are included at this stage. Perform simulations to verify correct functionality and conduct SNDR analysis. This will represent your best-case performance.

Ensure that a proper clocking scheme is implemented. If non-overlapping clocks are required, make sure you are able to generate and verify an appropriate non-overlapping clock version within your model.

Deliverables (due: 4/14/2026): Please submit a zipped Verilog-A library so that the instructional team can reproduce and re-run your simulations.

Phase III: Next, incorporate non-idealities into each functional block of your behavioral model. Include as many relevant non-ideal effects as possible and evaluate their impact systematically, one at a time.

Finally, provide a clear summary of your findings, highlighting how each non-ideality affects overall ADC performance.

Deliverables (due: 4/28/2026): Please submit a zipped Verilog-A library so that the instructional team can reproduce and re-run your simulations.

Phase IV: Replace one of the behavioral-level blocks with a transistor-level implementation and perform co-simulation. Verify both functionality and performance metrics, such as SNDR and ENOB. Finally, summarize your findings, including any discrepancies observed between the behavioral and transistor-level results.

Deliverables (due: 5/7/2026): A 4-page IEEE format report and a final presentation (~10 minutes).